Rising Waters

A Machine Learning Approach to Flood Prediction

Brainstorming & Idea Prioritization

Phase 1 — Ideation

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1. Introduction

This document captures the ideation and brainstorming process undertaken by the Rising Waters team to identify, evaluate, and prioritize the core problem and solution approach for the flood prediction system.

2. Problem Identification

Floods are among the most devastating natural disasters, responsible for thousands of deaths and millions of displacements annually. The team identified three root causes driving the problem:

- Lack of timely and accurate early-warning systems in flood-prone regions
- Conventional forecasting methods failing to incorporate multi-variable meteorological data
- Insufficient lead time for disaster management authorities to respond effectively

3. Brainstorming Session — Ideas Generated

3.1 Idea 1: Real-time sensor network

Deploy IoT water-level sensors across river systems and connect to a dashboard. Rejected at this stage due to high hardware cost and infrastructure requirements beyond project scope.

3.2 Idea 2: Satellite imagery analysis

Use remote sensing data to detect water coverage changes. Rejected due to data accessibility limitations and processing complexity for a short-duration project.

3.3 Idea 3: ML-based prediction from historical weather data

Train supervised classification models on historical rainfall and meteorological data to predict flood likelihood. Selected as the core approach — feasible, data-available, and directly addresses the early-warning gap.

3.4 Idea 4: Community reporting app

Crowdsource flood reports via a mobile app. Considered as a future enhancement, not the primary solution.

4. Idea Prioritization Matrix

Idea	Feasibility	Impact	Data Availability	Priority
ML prediction (historical data)	High	High	High	★★★★★
Real-time IoT sensors	Low	High	Medium	★★☆☆☆
Satellite imagery analysis	Medium	High	Low	★★★☆☆

Community reporting app	High	Medium	N/A	★★★☆☆

5. Selected Approach

The team selected Idea 3 — a machine learning-powered flood prediction system trained on historical weather data. The solution will:

- Use the Kaggle rainfall dataset (115 records, 10 meteorological features)
- Train and compare Decision Tree, Random Forest, KNN, and Gradient Boosting classifiers
- Deploy the best-performing model in a Flask web application
- Allow real-time input of weather parameters and return instant flood risk predictions

6. Conclusion

The brainstorming process identified a technically feasible, high-impact solution aligned with available data and the team's skill set. The machine learning approach addresses the core problem of timely flood prediction using proven classification algorithms.